

Survivorship: read the axis

What a survivorship curve is actually plotting, and why the same cohort can look like two different types depending on the y-axis.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/survivorship/

1 Before you open anything

Answer from what you already know. You are not expected to be right — the point is to have a number on paper to compare against.

PREDICTION

A survivorship curve is drawn as a perfectly straight line running from 1000 survivors per 1000 born at age class 0 down to 1 survivor at age class 10. How many of the original 1000 are still alive at age class 5, halfway to the maximum lifespan?

Write one sentence saying how you got that number.

2 Read the life table before you read the picture

Open the demonstration. The plot is on the left; the **Life table** panel on the right tabulates whichever curve its tabs point at. One age class is one tenth of the maximum lifespan.

1. Leave the buttons above the plot set to **Logarithmic y-axis**.
2. In the **Life table** panel, click the **Type II** tab.
3. Copy the listed rows out of the table columns **Alive $n(x)$** , **Surviving $l(x)$** and **Mortality $q(x)$** into the first table below.
4. Now click through the **Type I**, **Type II** and **Type III** tabs of the same panel and copy the four readout boxes sitting above the table into the second table below.
5. Return to the **Type II** tab and press **Linear y-axis** above the plot. Look at the plot, then look back at both tables you just filled in.

Age class x	Alive $n(x)$	Surviving $l(x)$	Mortality $q(x)$
0			
1			
2			
5			
9			
10			

Tab	Median lifespan (% of max lifespan)	Dies in age class 0 (%)	$q(x)$ range across life (%)	Bend off the straight log line (decades)
Type I				
Type II				
Type III				

a. Exactly one row has a bend of 0.00 decades. What is different about that row's $q(x)$ range compared with the other two?

b. Which numbers in either table changed when you pressed **Linear y-axis**?

c. In one sentence, describe what the Type II curve looks like on the linear axis, and name the type it now resembles.

The axis buttons change one thing only: how a survivor count is turned into a height on screen. If the picture changed and the life table did not, then what changed was the drawing, not the biology.

3 What makes a curve straight on a log axis

The panel **Your species — the three types are a continuum** draws a fourth curve whose shape you set. Moving the slider also points the life table and its four readouts at that curve, so the numbers follow you.

1. Set the axis back to **Logarithmic y-axis**.
2. Tick **Draw a fourth curve**.
3. Drag the **Mortality shape p** slider to its far left end, where the value beside it reads **p = 0.25**. Record the three readouts.
4. Repeat at **p = 0.50**, **p = 1.00**, **p = 2.00** and **p = 3.98**. The **Type II** button underneath the slider snaps you to exactly 1.00.

p	Median lifespan (% of max lifespan)	q(x) range across life (%)	Bend off the straight log line (decades)
0.25			
0.50			
1.00			
2.00			
3.98			

a. At which value of p is the bend exactly 0.00 decades? Write the q(x) range for that row, and say in one sentence what that range means about the fraction of the survivors dying in each age class.

b. The rows p = 0.50 and p = 2.00 report the same bend. Look at the two curves on screen: are they the same shape? State what the bend readout does measure and what it does not distinguish.

c. Complete the rule in your own words: a survivorship curve plots as a straight line on a logarithmic y-axis when —

d. Go back to your prediction on page 1. The Type II row for age class 5 in Part 2 is the answer the demonstration gives. If your prediction was near 500, say what you were assuming about equal distances on the y-axis; if it was not, say what you assumed instead.

4 Working backwards, and one shortcut that fails

A question rarely hands you p. It hands you a mortality figure and expects you to place the species. Keep **Draw a fourth curve** ticked and the axis on **Logarithmic y-axis**.

1. Move the **Mortality shape p** slider until **Dies in age class 0** reads as close to 70.0% as the slider allows. Record p, the bend, and the sentence printed under the slider.
2. Repeat until **Dies in age class 0** reads 25.0%.
3. Repeat until **Median lifespan** reads 32% of max lifespan.

Readout you are matching	p	Bend (decades)	What the sentence under the slider calls it
Dies in age class 0 = 70.0%			
Dies in age class 0 = 25.0%			
Median lifespan = 32% of max			

a. None of your three rows has a bend of 0.00 decades, so none of them is Type II. Two of the three sentences under the slider place the species *between* named types. What does that suggest about treating Type I, II and III as three boxes into which every species has to fall?

b. Press **Linear y-axis** and leave **Dashed straight-line reference** ticked. The dashed line runs from 1000 survivors at class 0 to 1 survivor at class 10, and no curve on screen follows it. A cohort that did follow it would lose the same *number* of individuals every class: $n(x) = 1000, 900, 800, \dots 200, 100$ for classes 0 through 9. Calculate $q(0)$ and $q(8)$ for that cohort, showing your working.

c. Using your two values from part b, explain where the shortcut “a straight survivorship line means a constant death rate” breaks down, and what has to be added to it to make it true.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it and check yourself — and if you were wrong, write down where the reasoning went off.

1. A cohort of 1000 loses exactly half of its survivors in every age class. Complete $n(x)$ for $x = 0$ to 5. How many individuals die during class 0, and how many during class 4? Both are the same $q(x)$ — explain in one sentence why the two counts differ so much.

2. A life table gives $n(x) = 1000, 60, 25, 14$ for age classes 0 to 3. Calculate $q(0)$, $q(1)$ and $q(2)$ to one decimal place. Is the proportion dying constant? Which of the three types is this cohort closest to, and would its log plot be straight?

3. A figure in a paper shows survivors falling as a straight line from 1000 down to 1 across ten age classes, but the y-axis has no scale printed on it. Explain why the straight line alone does not tell you the species is Type II, and state the one piece of information that would settle it. No calculation.

4. Take the demonstration's Type II cohort, which loses 49.9% of its survivors in every age class starting from 1000. Calculate the first age class at which fewer than 10 of the original 1000 are still alive. Show the working, not just the answer.

Problem 1 gives a clean 50% per class and problem 4 uses the demonstration's 49.9%. That difference is not a rounding slip. Every curve in the model is forced to end at exactly 1 survivor per 1000 at the maximum lifespan, which fixes the per-class factor at $1000^{1/10}$. Real cohorts are not forced to do anything of the kind.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Watch the **Mortality rate $q(x)$** bar chart below the plot while you click through the four life-table tabs. Then raise p slowly from 1.00 and find the point at which the bars stop looking level to your eye. Compare that p with what the bend readout was already reporting.
- The closing section notes that every curve is forced to end at exactly 1 survivor per 1000. Decide which of the four readouts would move if a species ended its maximum lifespan with 100 survivors instead, and which would not.
- Untick every curve except **Type III** and compare where it meets the bottom of the plot on each axis. The log axis cannot display zero survivors and the linear axis can — work out what that means for reading a real cohort that does reach zero.